

Testbench Environment Setup for RTL Design

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1 Introduction

This document presents the technical details of the SystemVerilog testbench developed to simulate a sigma-delta ADC followed by a CIC decimation filter. The testbench internally generates a sinusoidal or DC analog stimulus, adds flicker noise (1/f noise), models the integrator and comparator behavior of a first-order sigma-delta modulator, and captures filtered digital outputs. All stimulus and processing occur inside the testbench without external file dependencies. The structure enables clean waveform visualization in Vivado and provides a more realistic analog behavior through controlled noise injection.

2 Module Hierarchy

- **Filter_Top (sigma_delta_adc.sv)**
 - HF_Filter
 - * integrator : D_FF_N
 - * comb : HF_ROM
 - CIC Stages
 - * integrator : cic_integrator
 - * comb : cic_comb
 - fir_compensator

3 Input Signal Generation

3.1 Base Waveform

The input signal is generated as either a sinusoidal wave or a constant DC value. When sinusoidal mode is enabled, the input is defined as:

$$\text{adc_input}(t) = A \cdot \cos(2\pi ft) + \text{OFFSET} + n(t)$$

where $n(t)$ represents the filtered flicker noise.

- **Frequency:** 440 Hz
- **Amplitude:** 1.0 V
- **Offset:** 1.25 V
- **Clock Frequency:** 12.88 MHz

The noiseless waveform ranges approximately from 0.25 V to 2.25 V.

3.2 Sampling Duration

The total number of generated samples is:

$$\text{NUM_SAMPLES} = \text{NUM_CYCLES} \times \left(\frac{\text{BCLK}}{\text{FREQUENCY}} \right)$$

For 10 cycles of a 440 Hz waveform, this results in approximately 292,727 samples, corresponding to a simulation time of roughly 22.73 ms.

4 Flicker Noise Modeling

4.1 Raw Noise Generation

Uncorrelated white noise is produced using a scaled random function:

$$n_{\text{raw}} = \frac{\text{FLICKER_AMP} \cdot (r - 50)}{50}$$

where r is the integer output of the SystemVerilog `$random` function. This yields noise in the range $\pm \text{FLICKER_AMP}$, with a default value of ± 10 mV.

4.2 Low-Pass Filtering for Flicker Noise

To approximate $1/f$ behavior, a single-pole IIR filter is applied:

$$n(t) = 0.3 \cdot n(t-1) + 0.7 \cdot n_{\text{raw}}$$

This produces slow-varying drift characteristic of flicker noise.

4.3 Noise Range and SNR

- **FLICKER_AMP = 0.005:** ± 5 mV
- **FLICKER_AMP = 0.010:** ± 10 mV (default)
- **FLICKER_AMP = 0.020:** ± 20 mV

The approximate signal-to-noise ratio is:

$$\text{SNR} \approx 20 \log_{10} \left(\frac{1}{0.0058} \right) \approx 44.7 \text{ dB}$$

5 Input Waveform Behavior

Including flicker noise, the combined signal fluctuates slightly around the base sine wave. For example, at a sample corresponding to 180° phase:

$$\text{Base} = 0.25 \text{ V}, \quad n(t) \approx 0.003 \text{ V}, \quad \text{adc_input} \approx 0.253 \text{ V}$$

The overall statistical range observed during the simulation is:

$$0.243 \text{ V} \leq \text{adc_input}(t) \leq 2.257 \text{ V}$$

6 Sigma-Delta Modulator Modeling

6.1 Integrator Dynamics

The testbench includes a behavioral model of the RC integrator used in the sigma-delta feedback loop. The charge update on the internal node is approximated by:

$$\text{increase} = \frac{V_{CC} - v_n}{\text{CAP_FUDGE}}, \quad \text{decrease} = \frac{v_n}{\text{CAP_FUDGE}}$$

The integrator state v_n is updated as:

$$v_n(t+1) = \begin{cases} v_n(t) + \text{increase}, & \text{if } \text{adc_fb_pin} = 1 \\ v_n(t) - \text{decrease}, & \text{else} \end{cases}$$

6.2 Comparator Modeling

The LVDS input comparator is simply modeled as:

$$\text{adc_lvds_pin} = \begin{cases} 1, & \text{if } v_p > v_n \\ 0, & \text{otherwise} \end{cases}$$

7 CIC Decimation Output

The CIC decimator processes the modulator bitstream into filtered digital samples. The expected number of output samples is:

$$\frac{\text{NUM_SAMPLES}}{\text{OVERSAMPLE_RATE}} \approx \frac{292,727}{256} \approx 1,143$$

Output samples are displayed in the simulator console. Sample output entries include:

```
[Sample 100] Output: 0x7f4521  
[Sample 200] Output: 0x7f2340  
[Sample 300] Output: 0x7e5612
```

8 Recommended Waveform Visualization

To effectively visualize the system behavior in Vivado or GTKWave, monitor:

- `adc_input` (noisy analog waveform)
- `lvds_pin_p` and `lvds_pin_n` (integrator states)
- `adc_lvds_pin` (bitstream)
- `adc_output` (decimated output)
- `adc_valid` (sample strobe)

VCD dumping can be enabled by setting:

```
parameter DUMP_VCD = 1;
```

9 Test Variations

The following configurations can be used for exploratory testing:

- **DC Mode:** Produces constant offset voltage with noise.
- **High Noise Mode:** `FLICKER_AMP` increased to 0.05 for stress testing.
- **Frequency Sweep:** Frequency modified for bandwidth characterization.

10 Conclusion

This testbench provides a realistic and fully self-contained model of a sigma-delta ADC with CIC filtering. It eliminates external file dependencies, incorporates a practical flicker noise model, supports visualization in FPGA simulation tools, and produces clean, human-readable output suitable for analysis and documentation.